MATH 152 **QUIZ 10** NAME _____

Mrs. Bonny Tighe 25 points

11.10 – 11.11 TA_____Mon 11/19/2018

Due at the beginning of discussion on Monday 11/26/2018

1. Evaluate the limit using series $\lim_{x \to 0} \frac{\tan^{-1}(2x) - x}{x - \sin(2x)}$

2. Use the Binomial Theorem to expansion of the function as a power series. $f(x) = \sqrt[3]{8 + 2x^4}$

3. Find the MacLaurin series using a known series

a) $f(x) = 4\cos(2x^2)$ b) $f(x) = x^2 e^{4x}$

4. Find the Taylor series for $f(x) = \dfrac{1}{(3+x)^{3/2}}$ centered at $a = 1$

Find the radius of convergence.

5. Find the sum of the power series a) $\displaystyle\sum_{k=0}^{\infty} \frac{(-1)^k \pi^{2k}}{9^k (2k)!}$

b) $\sum_{k=0}^{\infty} \frac{(-1)^k \pi^{2k}}{3^{2k}(2k+1)!}$

6. Find the Taylor polynomial $T_4(x)$ for the function $f(x) = \sin x \ at \ a = 7\pi/6$ and use it to estimate $\sin(200\pi/180)$

7. Estimate $\cos(0.1)$ to three non-zero terms using the know MacLaurin series.

8. Use the **TAYLOR'S INEQUALITY** given below to Approximate the function $f(x) = \frac{1}{(1+3x)^3}$ by a Taylor polynomial of degree 3 at $a = 1$

How accurate is this approximation when $0.7 < x < 1.3$

If $\left|f^{(n+1)}(x)\right| \le M$ for $|x - a| \le d$, then the remainder $R_n(x) \le \frac{M}{(n+1)!}\left|x - a\right|^{n+1}$

for $|x - a| \le d$